

Esercizi Lez. 10 - Teoria

Esercizio 1

Sia $Z \sim N(0, 1)$ la normale standard. Calcolare:

$$P(Z > 0), \quad P(Z \geq 1.07), \quad P(Z < -1.45), \quad P(Z > -0.06), \quad P(Z \leq -2)$$

Risoluzione

$$\begin{aligned} 1. \quad P(Z > 0) &= 1 - P(Z \leq 0) = 1 - \Phi(0) \\ &= 1 - 0.5 = \boxed{0.5}. \end{aligned}$$

$$\begin{aligned} 2. \quad P(Z \geq 1.07) &= 1 - P(Z < 1.07) = 1 - \Phi(1.07) \\ &= 1 - 0.85769 = \boxed{0.14231}. \end{aligned}$$

$$\begin{aligned} 3. \quad P(Z < -1.45) &= \Phi(-1.45) = 1 - \Phi(1.45) \\ &= 1 - 0.92647 = \boxed{0.07353}. \end{aligned}$$

$$\begin{aligned} 4. \quad P(Z > -0.06) &= 1 - \Phi(-0.06) \\ &= 1 - (1 - \Phi(0.06)) \\ &= \Phi(0.06) = \boxed{0.52392}. \end{aligned}$$

$$\begin{aligned} 5. \quad P(Z \leq -2) &= \Phi(-2) = 1 - \Phi(2) \\ &= 1 - 0.97725 = \boxed{0.02275}. \end{aligned}$$

Esercizio 2

Sia $X \sim N(4, 25)$, calcolare:

$$P(X \leq 5), P(X \geq 2)$$

Risoluzione

Abbiamo $\mu = 4$ e $\sigma = 5$:

$$\begin{aligned} 1. \quad P(X \leq 5) &= \Phi\left(\frac{5 - \mu}{\sigma}\right) = \Phi\left(\frac{5 - 4}{5}\right) \\ &= \Phi(0.20) = \boxed{0.57926}. \end{aligned}$$

$$\begin{aligned} 2. \quad P(X \geq 2) &= 1 - \Phi\left(\frac{2 - 4}{5}\right) = 1 - \Phi\left(-\frac{2}{5}\right) = 1 - (1 - \Phi(0.40)) \\ &= \Phi(0.40) = \boxed{0.65542}. \end{aligned}$$

Esercizio 3

Determinare il 10^{mo} percentile di

$$Z \sim N(0, 1), \quad X \sim N(21, 3)$$

Risoluzione

1. Per $Z \sim N(0, 1)$:

$$\Phi(q_{0.10}) = 0.10$$

$$\Phi(-q_{0.10}) = 1 - 0.10 = 0.90 \approx 0.89973$$

$$-q_{0.10} = \Phi'(0.89973) \approx 1.28$$

$$q_{0.10} = \boxed{-1.28}.$$

2. Per $X \sim N(21, 3)$:

$$\Phi\left(\frac{q_{0.10} - 21}{\sqrt{3}}\right) = 0.10$$

$$\Phi\left(-\frac{q_{0.10} - 21}{\sqrt{3}}\right) = 1 - 0.10 = 0.90 \approx 0.89973$$

$$-\frac{q_{0.10} - 21}{\sqrt{3}} = \Phi'(0.89973) \approx 1.28$$

$$q_{0.10} = -1.28\sqrt{3} + 21 = \boxed{18.78}.$$
